

LMS Requirements Levels and Milestones

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Project Name: N/A

Project: Learning Management System (LMS)

Client Goal: Build a fully functional, scalable, and user-friendly LMS supporting admin/instructor/student workflows, gamification, assessments, certificates, analytics, AI support, and secure local + international payments.

1) Scope Summary

- ☐ Web-based LMS (responsive) with role-based access (Admin, Instructor, Student; optional: Content Editor, Support Agent).
- ☐ Core learning features: course catalog, enrollment, prerequisite gating, recorded/live classes, interactive assessments, certificates, analytics, and gamification.
- ☐ Integrations: payment gateways (local & international), email/SMS/WhatsApp notifications, optional Zoom/Google Meet/MS Teams for live classes, AI chatbot for student support.
- ☐ Deployment: cloud hosting with CI/CD, monitoring, backups, and basic analytics.

2) Modular Requirements

Core Platform

- ☐ User authentication & authorization (OAuth/email, optional SSO; password reset, 2FA optional).
- ☐ Role & permission management (Admin/Instructor/Student; granular module-level permissions).
- ☐ Organization settings (brand, logo, theme, timezone, email/SMS settings).
- ☐ CMS-lite: pages (Home, About, FAQ, Contact), banners, blog/announcements.
- ☐ Search & filters for courses/instructors (category, level, language, price, rating, tags).

Admin Panel

- ☐ Full control over courses, lessons, modules, and catalogs.
- ☐ User management (create/disable roles; bulk import; KYC if needed).
- ☐ Payment/Orders: pricing, coupons, refunds, invoices, tax/VAT settings.
- ☐ System settings: gateway keys, integrations, email/SMS templates, notification rules.
- ☐ Moderation (reviews, Q&A, reports, content flags).
- ☐ Audit logs & activity history.

Instructor Panel

- ☐ Course builder (modules, lessons: video, PDF, SCORM-lite, external links, assignments).
- ☐ Content scheduling (drip), prerequisite mapping, and visibility controls.
- ☐ Student management (roster, enrollment status, communication tools).
- ☐ Assessment tools: quizzes (MCQ/True-False/Short answer), polls, challenges.
- ☐ Grading & rubric management; manual override; feedback on submissions.
- ☐ Analytics: per-course performance, engagement heatmaps, quiz item analysis.
- ☐ Payout dashboard (if revenue sharing/marketplace), statements & export.

Student Panel

- ☐ Dashboard: enrolled courses, progress tracking, deadlines, certificates, rewards.
- ☐ Course player: playlist, notes, bookmarks, discussion/Q&A, and resources download.
- ☐ Prerequisite guardrails (cannot access paid course until required course completion).
- ☐ Assessments: take quizzes/polls/challenges; instant feedback where applicable.
- ☐ Certificates: view, download, share; verification link/QR.
- ☐ Payments: checkout (local + international), invoices, and refund requests.
- ☐ AI chatbot: FAQs, course guidance, study tips, and quick answers from the knowledge base.

Content & Delivery

- ☐ Recorded classes: upload/stream (VOD), adaptive bitrate via secure video hosting (e.g., Vimeo/Cloudflare Stream/YouTube Unlisted as interim).
- ☐ Live class suggestions & scheduling (Zoom/Google Meet/MS Teams integration; calendar invites; attendance logs).
- ☐ File/resource library with versioning; size/type restrictions.
- ☐ Discussion/Q&A per lesson; reaction/emojis; moderation.

Gamification

- ☐ Points, badges, leaderboards, streaks, progress rewards, achievement milestones.
- ☐ Rule engine: configurable triggers (e.g., lesson complete, quiz score, streak days).

- ☐ Anti-gaming controls (cooldowns, unique completion checks).
- ☐ Public/Private leaderboards by course/category.

Assessments

- ☐ Quiz engine (question bank, randomization, time limits, negative marking optional).
- ☐ Polls & challenges (timed, peer leaderboard optional).
- ☐ Attempts & grading policy (best of N, last attempt, pass marks).
- ☐ Item analysis, export, and regrade workflow.

Certificates & Verification

- ☐ Template editor (logo, signature, fields, QR/unique ID).
- ☐ Auto-issue on completion rules; manual issue by instructor/admin.
- ☐ Public verification page (enter code or scan QR to validate certificate).

Payments & Commerce

- ☐ Secure checkout with local + international gateways (e.g., SSLCommerz/bKash/Nagad + Stripe/PayPal).
- ☐ One-time purchase, bundles, coupons, gift codes; future-ready for subscriptions.
- ☐ Invoices/receipts (download PDF), basic tax/VAT support, refund workflow.
- ☐ Webhooks & payment reconciliation; order management.

Communication & Notifications

- ☐ Email/SMS/WhatsApp notifications (enrollment, reminders, certificate issued, payment status).
- ☐ In-app notifications & announcement center.
- ☐ Message center: instructor ↔ student (optional moderation/filters).

AI Chatbot Integration

- ☐ Knowledge base ingestion (FAQs, policy pages, course outlines).
- ☐ Contextual Q&A (course progress aware, but privacy-safe).
- ☐ Triage to human support (ticket handoff) when needed.
- ☐ Admin controls: intents, answer tone, escalation rules, and logs.

Reporting & Analytics

- ☐ Dashboards: revenue, enrollments, completion rates, engagement, quiz outcomes.
- ☐ Instructor reports (per course, cohort analysis, student at-risk alerts).
- ☐ Student-level transcript/activity exports (CSV/XLSX).

Security, Compliance & DevOps

- ☐ OWASP-aligned security; RBAC; CSRF/XSS mitigations; rate limiting.
- ☐ Media security (signed URLs, domain-level restrictions, watermarking optional).
- ☐ GDPR-style privacy controls (consent, data export/delete on request).
- ☐ Backups, monitoring/alerts, error tracking, uptime SLO (e.g., 99.5%).

3) Timeline & Milestones (Indicative 5-Week Plan)

Project Start (T0): Once SOW is signed & access/credentials are shared.

Phase	Duration	Days	Key Deliverables
0. WireFrame & UI/UX Design	5 days	D1-D5	Requirement freeze, wireframes, UI kit
1. Core Platform & Auth	4 days	D6-D9	User roles (Admin/Instructor/Student), login/signup, settings
2. Course Builder + Content Delivery	5 days	D10-D14	Course/lesson CRUD, recorded class upload, basic live class scheduling
3. Student Dashboard & Course Player	4 days	D15-D18	Progress tracking, prerequisite guardrails, discussion/Q&A
4. Assessment & Gamification (MVP)	5 days	D19-D23	Quizzes, polls, points, badges, leaderboards (basic rules)
5. Certificates & Payments	5 days	D24-D28	Certificate generator + QR, local+international gateway integration
6. AI Chatbot & Analytics	2 days	D29-D30	KB-based chatbot, admin/instructor reports
7. Testing & QA	3 days	D31-D33	Bug fixes, security,

Phase

performance

8. Deployment
Phase

2 day

D34-D35

Production deploy,
training, docs

Milestones

- ☐ M1 (D5): Design sign-off (UI kit, wireframes)
- ☐ M2 (D14): Courses & content delivery working
- ☐ M3 (D18): Student experience (dashboard + player + prereqs) ready
- ☐ M4 (D23): Quizzes & gamification active
- ☐ M5 (D28): Certificates + Payments live
- ☐ M6 (D30): AI chatbot MVP + analytics dashboard
- ☐ M7 (D35): Production launch & handover

